Local Interpretability Report

Customer ID: 100005

Default Probability (TARGET=1): 0.5560

Decision explanation

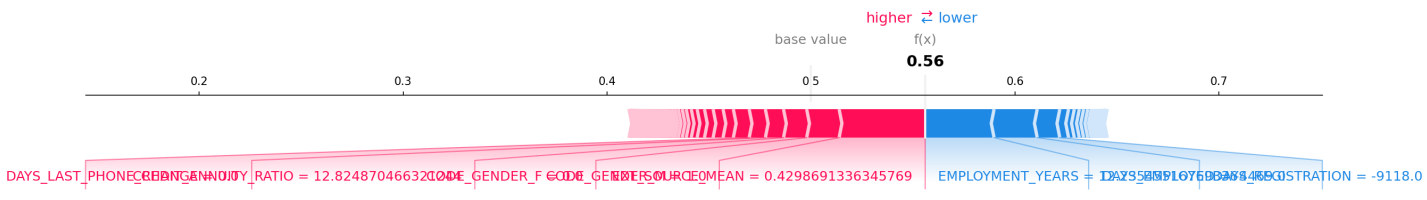
The model predicts default with default probability 55.60%. The class decision uses a 0.50 probability threshold (ties go to class 0). The SHAP baseline default probability is 50.00%; positive contributions increase default risk and negative contributions reduce it. Strongest increases in risk: EXT_SOURCE_MEAN (+4.22 percentage points); CODE_GENDER_M (+1.59 percentage points); CODE_GENDER_F (+1.13 percentage points). Strongest reductions in risk: EMPLOYMENT_YEARS (-3.38 percentage points); DAYS_EMPLOYED (-2.12 percentage points); DAYS_REGISTRATION (-1.11 percentage points). The test client's outcome is unknown, so correctness cannot be assessed.

Largest SHAP contributions to default probability

Feature	Client value	Contribution (percentage points)
EXT_SOURCE_MEAN	0.4299	+4.22
EMPLOYMENT_YEARS	12.2355	-3.38
DAYS_EMPLOYED	-4469.0000	-2.12
CODE_GENDER_M	1.0000	+1.59
CODE_GENDER_F	0.0000	+1.13
DAYS_REGISTRATION	-9118.0000	-1.11
CREDIT_ANNUITY_RATIO	12.8249	+0.89
DAYS_LAST_PHONE_CHANGE	0.0000	+0.79

Interactive Plotly client and population charts: client_test.html

SHAP Force Plot



Client Data vs. Population Average

Feature	Client Value	Dataset Average
CNT_CHILDREN	0.00	0.40
AMT_GOODS_PRICE	180000.00	462618.84
REGION_POPULATION_RELATIVE	0.04	0.02
DAYS_EMPLOYED	-4469.00	-2319.06
DAYS_REGISTRATION	-9118.00	-4967.65
DAYS_ID_PUBLISH	-1623.00	-3051.71
FLAG_MOBIL	1.00	1.00
FLAG_EMP_PHONE	1.00	0.81
FLAG_WORK_PHONE	0.00	0.20
FLAG_CONT_MOBILE	1.00	1.00
FLAG_PHONE	0.00	0.26
FLAG_EMAIL	0.00	0.16
CNT_FAM_MEMBERS	2.00	2.15
REGION_RATING_CLIENT	2.00	2.04
REGION_RATING_CLIENT_W_CITY	2.00	2.01
HOUR_APPR_PROCESS_START	9.00	12.01
REG_REGION_NOT_LIVE_REGION	0.00	0.02
REG_REGION_NOT_WORK_REGION	0.00	0.06
LIVE_REGION_NOT_WORK_REGION	0.00	0.04
REG_CITY_NOT_LIVE_CITY	0.00	0.08
REG_CITY_NOT_WORK_CITY	0.00	0.22

LIVE_CITY_NOT_WORK_CITY	0.00	0.17
APARTMENTS_AVG	0.09	0.11
BASEMENTAREA_AVG	0.08	0.08
YEARS_BEGINEXPLUATATION_AVG	0.98	0.98
ELEVATORS_AVG	0.00	0.04
ENTRANCES_AVG	0.14	0.15
FLOORSMAX_AVG	0.17	0.20
LANDAREA_AVG	0.05	0.06
LIVINGAREA_AVG	0.07	0.09
NONLIVINGAREA_AVG	0.00	0.02
APARTMENTS_MODE	0.08	0.10
BASEMENTAREA_MODE	0.07	0.08
YEARS_BEGINEXPLUATATION_MODE	0.98	0.98
ELEVATORS_MODE	0.00	0.04
ENTRANCES_MODE	0.14	0.14
FLOORSMAX_MODE	0.17	0.20
LANDAREA_MODE	0.05	0.05
LIVINGAREA_MODE	0.07	0.09
NONLIVINGAREA_MODE	0.00	0.01
APARTMENTS_MEDI	0.09	0.11
BASEMENTAREA_MEDI	0.08	0.08
YEARS_BEGINEXPLUATATION_MEDI	0.98	0.98

ELEVATORS_MEDI	0.00	0.04
ENTRANCES_MEDI	0.14	0.14
FLOORSMAX_MEDI	0.17	0.20
LANDAREA_MEDI	0.05	0.06
LIVINGAREA_MEDI	0.07	0.09
NONLIVINGAREA_MEDI	0.00	0.02
TOTALAREA_MODE	0.07	0.09
OBS_30_CNT_SOCIAL_CIRCLE	0.00	1.45
DEF_30_CNT_SOCIAL_CIRCLE	0.00	0.14
OBS_60_CNT_SOCIAL_CIRCLE	0.00	1.43
DEF_60_CNT_SOCIAL_CIRCLE	0.00	0.10
DAYS_LAST_PHONE_CHANGE	0.00	-1077.77
FLAG_DOCUMENT_2	0.00	0.00
FLAG_DOCUMENT_3	1.00	0.79
FLAG_DOCUMENT_4	0.00	0.00
FLAG_DOCUMENT_5	0.00	0.01
FLAG_DOCUMENT_6	0.00	0.09
FLAG_DOCUMENT_7	0.00	0.00
FLAG_DOCUMENT_8	0.00	0.09
FLAG_DOCUMENT_9	0.00	0.00
FLAG_DOCUMENT_10	0.00	0.00
FLAG_DOCUMENT_11	0.00	0.00

FLAG_DOCUMENT_12	0.00	0.00
FLAG_DOCUMENT_13	0.00	0.00
FLAG_DOCUMENT_14	0.00	0.00
FLAG_DOCUMENT_15	0.00	0.00
FLAG_DOCUMENT_16	0.00	0.00
FLAG_DOCUMENT_17	0.00	0.00
FLAG_DOCUMENT_18	0.00	0.00
FLAG_DOCUMENT_19	0.00	0.00
FLAG_DOCUMENT_20	0.00	0.00
FLAG_DOCUMENT_21	0.00	0.00
AMT_REQ_CREDIT_BUREAU_HOUR	0.00	0.00
AMT_REQ_CREDIT_BUREAU_DAY	0.00	0.00
AMT_REQ_CREDIT_BUREAU_WEEK	0.00	0.00
AMT_REQ_CREDIT_BUREAU_MON	0.00	0.01
AMT_REQ_CREDIT_BUREAU_QRT	0.00	0.48
AMT_REQ_CREDIT_BUREAU_YEAR	3.00	1.86
AGE_YEARS	49.46	43.99
EMPLOYMENT_YEARS	12.24	6.35
CREDIT_INCOME_RATIO	2.25	3.17
ANNUITY_INCOME_RATIO	0.18	0.18
CREDIT_ANNUITY_RATIO	12.82	17.18
INCOME_PER_PERSON	49500.00	98078.81

EXT_SOURCE_MEAN	0.43	0.51
NAME_CONTRACT_TYPE_Cash loans	1.00	0.99
NAME_CONTRACT_TYPE_Revolving loans	0.00	0.01
CODE_GENDER_F	0.00	0.67
CODE_GENDER_M	1.00	0.33
CODE_GENDER_XNA	0.00	0.00
FLAG_OWN_CAR_N	1.00	0.66
FLAG_OWN_CAR_Y	0.00	0.34
FLAG_OWN_REALTY_N	0.00	0.31
FLAG_OWN_REALTY_Y	1.00	0.69
NAME_TYPE_SUITE_Children	0.00	0.01
NAME_TYPE_SUITE_Family	0.00	0.12
NAME_TYPE_SUITE_Group of people	0.00	0.00
NAME_TYPE_SUITE_Missing	0.00	0.02
NAME_TYPE_SUITE_Other_A	0.00	0.00
NAME_TYPE_SUITE_Other_B	0.00	0.00
NAME_TYPE_SUITE_Spouse, partner	0.00	0.03
NAME_TYPE_SUITE_Unaccompanied	1.00	0.82
NAME_INCOME_TYPE_Businessman	0.00	0.00
NAME_INCOME_TYPE_Commercial associate	0.00	0.23

NAME_INCOME_TYPE_Maternity leave	0.00	0.00
NAME_INCOME_TYPE_Pensioner	0.00	0.19
NAME_INCOME_TYPE_State servant	0.00	0.07
NAME_INCOME_TYPE_Student	0.00	0.00
NAME_INCOME_TYPE_Unemployed	0.00	0.00
NAME_INCOME_TYPE_Working	1.00	0.50
NAME_EDUCATION_TYPE_Academic degree	0.00	0.00
NAME_EDUCATION_TYPE_Higher education	0.00	0.26
NAME_EDUCATION_TYPE_Incomplete higher	0.00	0.04
NAME_EDUCATION_TYPE_Lower secondary	0.00	0.01
NAME_EDUCATION_TYPE_Secondary / secondary special	1.00	0.70
NAME_FAMILY_STATUS_Civil marriage	0.00	0.09
NAME_FAMILY_STATUS_Married	1.00	0.66
NAME_FAMILY_STATUS_Separated	0.00	0.06
NAME_FAMILY_STATUS_Single / not married	0.00	0.14
NAME_FAMILY_STATUS_Unknown	0.00	0.00
NAME_FAMILY_STATUS_Widow	0.00	0.05
NAME_HOUSING_TYPE_Co-op apartment	0.00	0.00

NAME_HOUSING_TYPE_House / apartment	1.00	0.90
NAME_HOUSING_TYPE_Municipal apartment	0.00	0.03
NAME_HOUSING_TYPE_Office apartment	0.00	0.01
NAME_HOUSING_TYPE_Rented apartment	0.00	0.01
NAME_HOUSING_TYPE_With parents	0.00	0.05
OCCUPATION_TYPE_Accountants	0.00	0.03
OCCUPATION_TYPE_Cleaning staff	0.00	0.01
OCCUPATION_TYPE_Cooking staff	0.00	0.02
OCCUPATION_TYPE_Core staff	0.00	0.09
OCCUPATION_TYPE_Drivers	0.00	0.06
OCCUPATION_TYPE_HR staff	0.00	0.00
OCCUPATION_TYPE_High skill tech staff	0.00	0.04
OCCUPATION_TYPE_IT staff	0.00	0.00
OCCUPATION_TYPE_Laborers	0.00	0.18
OCCUPATION_TYPE_Low-skill Laborers	1.00	0.01
OCCUPATION_TYPE_Managers	0.00	0.07
OCCUPATION_TYPE_Medicine staff	0.00	0.03
OCCUPATION_TYPE_Missing	0.00	0.32
OCCUPATION_TYPE_Private service staff	0.00	0.01

OCCUPATION_TYPE_Realty agents	0.00	0.00
OCCUPATION_TYPE_Sales staff	0.00	0.10
OCCUPATION_TYPE_Secretaries	0.00	0.00
OCCUPATION_TYPE_Security staff	0.00	0.02
OCCUPATION_TYPE_Waiters/barmen staff	0.00	0.00
WEEKDAY_APPR_PROCESS_START_FRIDAY	1.00	0.15
WEEKDAY_APPR_PROCESS_START_MONDAY	0.00	0.17
WEEKDAY_APPR_PROCESS_START_SATURDAY	0.00	0.09
WEEKDAY_APPR_PROCESS_START_SUNDAY	0.00	0.04
WEEKDAY_APPR_PROCESS_START_THURSDAY	0.00	0.17
WEEKDAY_APPR_PROCESS_START_TUESDAY	0.00	0.20
WEEKDAY_APPR_PROCESS_START_WEDNESDAY	0.00	0.17
ORGANIZATION_TYPE_Advertising	0.00	0.00
ORGANIZATION_TYPE_Agriculture	0.00	0.01
ORGANIZATION_TYPE_Bank	0.00	0.01
ORGANIZATION_TYPE_Business Entity Type 1	0.00	0.02
ORGANIZATION_TYPE_Business Entity Type 2	0.00	0.03
ORGANIZATION_TYPE_Business Entity Type 3	0.00	0.22
ORGANIZATION_TYPE_Cleaning	0.00	0.00
ORGANIZATION_TYPE_Construction	0.00	0.02

ORGANIZATION_TYPE_Culture	0.00	0.00
ORGANIZATION_TYPE_Electricity	0.00	0.00
ORGANIZATION_TYPE_Emergency	0.00	0.00
ORGANIZATION_TYPE_Government	0.00	0.03
ORGANIZATION_TYPE_Hotel	0.00	0.00
ORGANIZATION_TYPE_Housing	0.00	0.01
ORGANIZATION_TYPE_Industry: type 1	0.00	0.00
ORGANIZATION_TYPE_Industry: type 10	0.00	0.00
ORGANIZATION_TYPE_Industry: type 11	0.00	0.01
ORGANIZATION_TYPE_Industry: type 12	0.00	0.00
ORGANIZATION_TYPE_Industry: type 13	0.00	0.00
ORGANIZATION_TYPE_Industry: type 2	0.00	0.00
ORGANIZATION_TYPE_Industry: type 3	0.00	0.01
ORGANIZATION_TYPE_Industry: type 4	0.00	0.00
ORGANIZATION_TYPE_Industry: type 5	0.00	0.00
ORGANIZATION_TYPE_Industry: type 6	0.00	0.00
ORGANIZATION_TYPE_Industry: type 7	0.00	0.00
ORGANIZATION_TYPE_Industry: type 8	0.00	0.00

ORGANIZATION_TYPE_Industry: type 9	0.00	0.01
ORGANIZATION_TYPE_Insurance	0.00	0.00
ORGANIZATION_TYPE_Kindergarten	0.00	0.02
ORGANIZATION_TYPE_Legal Services	0.00	0.00
ORGANIZATION_TYPE_Medicine	0.00	0.04
ORGANIZATION_TYPE_Military	0.00	0.01
ORGANIZATION_TYPE_Mobile	0.00	0.00
ORGANIZATION_TYPE_Other	0.00	0.06
ORGANIZATION_TYPE_Police	0.00	0.01
ORGANIZATION_TYPE_Postal	0.00	0.01
ORGANIZATION_TYPE_Realtor	0.00	0.00
ORGANIZATION_TYPE_Religion	0.00	0.00
ORGANIZATION_TYPE_Restaurant	0.00	0.01
ORGANIZATION_TYPE_School	0.00	0.03
ORGANIZATION_TYPE_Security	0.00	0.01
ORGANIZATION_TYPE_Security Ministries	0.00	0.01
ORGANIZATION_TYPE_Self-employed	1.00	0.12
ORGANIZATION_TYPE_Services	0.00	0.01
ORGANIZATION_TYPE_Telecom	0.00	0.00
ORGANIZATION_TYPE_Trade: type 1	0.00	0.00

ORGANIZATION_TYPE_Trade: type 2	0.00	0.00
ORGANIZATION_TYPE_Trade: type 3	0.00	0.01
ORGANIZATION_TYPE_Trade: type 4	0.00	0.00
ORGANIZATION_TYPE_Trade: type 5	0.00	0.00
ORGANIZATION_TYPE_Trade: type 6	0.00	0.00
ORGANIZATION_TYPE_Trade: type 7	0.00	0.03
ORGANIZATION_TYPE_Transport: type 1	0.00	0.00
ORGANIZATION_TYPE_Transport: type 2	0.00	0.01
ORGANIZATION_TYPE_Transport: type 3	0.00	0.00
ORGANIZATION_TYPE_Transport: type 4	0.00	0.02
ORGANIZATION_TYPE_University	0.00	0.00
ORGANIZATION_TYPE_XNA	0.00	0.19
HOUSETYPE_MODE_Missing	1.00	0.48
HOUSETYPE_MODE_block of flats	0.00	0.51
HOUSETYPE_MODE_specific housing	0.00	0.01
HOUSETYPE_MODE_terraced house	0.00	0.00
WALLSMATERIAL_MODE_Block	0.00	0.03
WALLSMATERIAL_MODE_Missing	1.00	0.49

WALLSMATERIAL_MODE_Mixed	0.00	0.01
WALLSMATERIAL_MODE_Monolithic	0.00	0.01
WALLSMATERIAL_MODE_Others	0.00	0.01
WALLSMATERIAL_MODE_Panel	0.00	0.23
WALLSMATERIAL_MODE_Stone, brick	0.00	0.21
WALLSMATERIAL_MODE_Wooden	0.00	0.02
EMERGENCYSTATE_MODE_Missing	1.00	0.46
EMERGENCYSTATE_MODE_No	0.00	0.54
EMERGENCYSTATE_MODE_Yes	0.00	0.01